LESSON 02 · MISSION GUIDE

The Bonus Run

Teach Python to make decisions, and to repeat itself without copy-paste.

From **Siobhan, Reward & Benefits**
To **Jen**
Subject **Bonus run (the napkin)**

Hi Jen. Gerald has written this year's bonus rules on a napkin. An actual napkin. It's pinned to the board next to the fire exit map: nothing under 12 months of service, 8 % for a rating of 4 or 5, 4 % for a 3, nothing below.

Simple enough for a person. Not for MACRO-9000, which in 2019 paid everyone, including Biscuit, and produced a total of minus forty pounds. Dave called it "a rounding thing". I need the same decision for all 200 people, every time, and a total that is the total. Mo says that's called "a loop". — S.

What you'll be able to do after today

- [ ] Make a script choose between outcomes with `if`, `elif` and `else`
- [ ] Repeat a block for every item with `for`, and a fixed number of times with `range()`
- [ ] Repeat until something is true with `while`, and stop a runaway loop
- [ ] Keep a running total, a counter and a "best so far" across a loop
- [ ] Skip an item with `continue`, and read indentation the way Python reads it

Why this matters at work

- **Every policy is an `if`.** Bonus rules, holiday entitlement, overtime bands: "if this, then that, otherwise...". As code, the rule is applied identically to everyone.
- **"Drag the formula down" is a loop.** It doesn't care whether there are 5 rows or 50,000, and it can't be dragged one row short.

Four big ideas

1. Indentation is the structure	VBA has `End If` and `Next`. Python has a colon and 4 spaces: the indented lines belong to the `if` or `for`. Un-indent, and you're outside it.
2. First true wins	In an `if / elif / else` chain Python runs the *first* true branch and skips the rest. Put the strictest check first.
3. `for` hands you one item at a time	`for name in names:` is `For Each`. One item per trip; you never write `i = i + 1`.
4. Set up before, finish after	A running total starts at 0 *before* the loop, grows *inside*, is printed *after*. Put the `= 0` inside and it resets every trip (ask Dave).

Three habits that will bite you

- Forgetting the **colon** at the end of `if`, `elif`, `else`, `for`, `while` lines.
- Writing `=` in a condition. `if rating = 3:` is a syntax error; you mean `==`.
- A `while` loop whose condition never becomes false. `Ctrl+C` stops it. Then fix the loop.

How today works

15 MIN
Lecture
Lars runs the napkin rules live, then loops them

60 MIN
Exercise sheet
Seven tickets. You write every file yourself

15 MIN
Dave's Leftovers #2
The bonus run that paid Biscuit

The inbox

#	FROM	TICKET	YOU'LL USE
1	Siobhan	One bonus decision	`if / elif / else`
2	Siobhan	The bonus run, for everyone	`for`, running total
3	Priya	Twelve months of pension	`range()`, running total
4	Gerald	Salary bands	`elif` chain, counters, "highest so far"
5	Mo	How many years to 35k?	`while`
6	Priya	Holiday allowance	`continue`, `+=`
7 ⭐	Priya	Dave's payroll calendar	everything, plus `%`

You're done when...

- [ ] `my_work/` has `ex1` to `ex6` (and `ex7` if you're feeling brave), each printing the expected output
- [ ] Dave's bonus run pays the right people and the total is £3,940.00, with one commit per bug
- [ ] Everything is pushed to `lesson-02` and you've opened a pull request



Next lesson: Dave's Spaghetti. You'll notice today that you typed the bonus rule twice. Dave typed it forty-one times. Next time you'll write it once, give it a name, and call it whenever you like: that's a function.